SYSTEMS BIOLOGY AND NETWORK MODELLING ASSIGNMENT

VEDAM GAYATHRI, SE23UCAB021

Part 1.1 — Primary paper

Paper: node2vec: Scalable Feature Learning for Networks

Authors: Aditya Grover and Jure Leskovec

Part 1.2 — Synopsis

1. Question

Many network analysis tasks, such as node classification and link prediction, require numerical features that represent the nodes and their relationships. Traditionally, these features are manually engineered using network properties, which can be time-consuming and may not generalize across different prediction tasks. Existing network representation methods also have limitations because they use relatively rigid neighbourhood sampling strategies. However, real-world networks exhibit different connectivity patterns, including homophily (nodes belonging to similar communities) and structural equivalence (nodes sharing similar structural roles). The authors address the problem of developing a scalable and flexible method that can learn meaningful low-dimensional representations of network nodes while capturing these different connectivity patterns.

2. Method

The authors propose node2vec, a framework that learns continuous feature representations of nodes using biased second-order random walks. The algorithm represents each node as a low-dimensional numerical vector and aims to preserve its network neighborhood. It uses two parameters, p and q, to control random-walk exploration. The return parameter (p) controls the likelihood of revisiting the previous node, while the in-out parameter (q) controls whether the walk explores locally or moves towards more distant nodes. Higher q values encourage breadth-first search (BFS)-like exploration, while lower q values encourage depth-first search (DFS)-like exploration. The generated node sequences are used as input to a Skip-gram model, which learns node embeddings by predicting neighboring nodes. The optimization technique uses stochastic gradient descent and negative sampling. The method assumes conditional independence of neighborhood nodes given the source-node representation and symmetry between source

and neighborhood nodes in the feature space. The learned node representations can also be combined using binary operators to generate edge representations for link prediction.

3. Key result

The authors evaluated node2vec on multi-label classification and link prediction using real-world networks, including social networks, information networks and a human protein–protein interaction network. The results showed that node2vec outperformed existing approaches such as DeepWalk and LINE in several experimental settings. It achieved improvements of up to 26.7% in multi-label classification and 12.6% in link prediction compared with the relevant state-of-the-art and heuristic baselines, respectively. The experiments also demonstrated that different random-walk strategies capture different network structures. Furthermore, the method showed robustness to missing and noisy edges and scalability to networks containing up to one million nodes. These findings support the importance of flexible neighbourhood exploration for learning useful network representations.

4. Limitation

One limitation is the dependence on parameters such as p and q, since different networks and prediction tasks may require different exploration strategies. Selecting appropriate parameters can introduce additional computational overhead. Furthermore, the method relies on assumptions of conditional independence and symmetry in feature space, which may not fully represent the complex dependencies present in real-world networks. Although the authors demonstrate strong predictive performance, the learned embeddings do not directly explain the biological meaning of individual features. The authors also identify the interpretability of edge representations, particularly the effectiveness of the Hadamard operator, as an area for future investigation. These limitations suggest that node2vec is useful for learning network representations, but its predictions and parameter choices require careful evaluation.